

Response to Reviewers

MIDS: Detecting Stealthy Masquerade and Tampering Attacks on CAN Bus via Bidirectional Mamba

Previous Manuscript ID: T-IFS-21023-2025

Current Manuscript ID: T-IFS-25937-2026

Submission: Resubmission after Reject decision of 16-Jul-2025

Dear Editor and Reviewers,

We thank the editor and the three reviewers for the rigorous and constructive feedback on our previous submission. The revised manuscript addresses every concern raised; this letter maps each reviewer comment to the corresponding revision and indicates the section, table, or figure of the revised paper where the change can be inspected.

The four most substantive revisions are:

- A **new cross-dataset benchmark** (**Table VII, Section VI-F**) in which all eight reproduced baselines are re-trained from scratch on four public datasets (OTIDS, ROAD, CrySyS, CT&T) plus our Tesla Model 3 dataset, under a single unified protocol with 5-fold block-shuffled cross-validation. A total of **160** runs underpin the new table.
- A **formal threat model and attack taxonomy** (**Section IV**) that adopts the Cho–Shin masquerade/fabrication naming convention and explicitly describes attacker capabilities, in-situ substitution semantics, hardware-assisted CRC bypass, and the offline synthesis of evaluation traces.
- Two new subsections on **data-leakage prevention and the evaluation protocol** (**Section VI-A4 and VI-A5**), disclosing the block-shuffled 5-fold split, the L -frame buffer, the cross-scenario interleaving, and an empirical scenario-leakage verification.
- A new **Computational Efficiency and Deployability** section (**Section VI-G, Table VI**) reporting parameters, FLOPs, single-window latency, and peak memory for MIDS and the strongest baselines.

In the rest of this letter, comments are quoted verbatim in italics and our responses follow in blue. Cross-references to sections, tables, and figures point to the *revised* manuscript.

Response to Reviewer 1

We thank Reviewer 1 for the supportive and detailed feedback. The seven concerns are addressed below.

Reviewer comment. *The paper lacks a proper description of an attacker model. It is unclear what tampering capabilities the attackers possess and what specific modifications in the CAN signals the IDS targets.*

Response. We have added **Section IV** (“Threat Model and Attack Taxonomy”), a dedicated section formalising the attacker model. **Section IV-A** enumerates three attacker capabilities (Software-level Control, In-situ Modification, Hardware-assisted CRC bypass). **Section IV-B** catalogues the three concrete attack vectors (ID-only masquerade, Data-only tampering, Combined masquerade) with worked Tesla examples (e.g., substituting the door-status ID 0x102 with the steering-angle ID 0x132). **Section IV-C** (“Implications for Detection”) then derives the two design requirements (frequency-invariance robustness and cross-field reasoning) that motivate MIDS’s dual-stream architecture.

Reviewer comment. *Recent literature commonly uses a specific naming convention for attacks to avoid confusion. I recommend that the authors adopt this naming convention (Cho & Shin, 2016).*

Response. The Cho–Shin convention (Cho and Shin, USENIX Security 2016) is now adopted throughout the manuscript. Specifically, **Section IV** opens by explicitly distinguishing *fabrication* attacks (frames appended to the bus, disturbing per-ID periodicity) from *masquerade* attacks (legitimate frame suppressed and substituted in situ, preserving periodicity). All three attack vectors considered in this paper are instances of the masquerade pattern, which we make explicit in the opening paragraph of **Section IV** and in the dataset-synthesis discussion in **Section VI-A3**.

Reviewer comment. *There are significant issues with the presentation of the attacks. [...] Two of the presented tampering strategies modify the ID field, effectively turning these tampering attacks into injection attacks. This presents a significant contradiction to the paper’s original positioning.*

Response. This was a critical misunderstanding caused by our earlier writing, which we have now corrected. Under the Cho–Shin taxonomy adopted in the revised **Section IV**, modifying the ID field is *not* equivalent to an injection attack: an injection attack appends an extra frame to the bus (raising the per-ID frame rate), whereas our ID-only masquerade *substitutes* a legitimate frame at its original transmission slot (preserving the per-ID rate). We have rewritten the opening paragraph of **Section IV** to make this in-situ semantics explicit, and **Section IV-B** now describes all three masquerade variants under this consistent framing.

Reviewer comment. *Another serious problem is that the evaluation lacks any comparison with prior work. [...] They should run their approach on at least some of these datasets to enable a comparison of results.*

Response. This is now the central new contribution of the revision. **Section VI-F** reports MIDS’s performance on four public datasets (OTIDS, ROAD, CrySyS, CT&T) using **Table VI** (per-metric MIDS breakdown). In addition, **Table VII** (“Cross-dataset F1 (%) of MIDS against eight baselines under the unified 5-fold protocol”) reports F1 for *all* eight reproduced baselines on *all* five datasets, totalling 160 runs. The baseline reproduction procedure is documented in **Section VI-B3** (“Baseline Reproduction Protocol”); the full code is publicly released.

Reviewer comment. *Since the proposed method would be deployed in a resource-constrained embedded environment, the paper should include a brief discussion of the method’s resource requirements.*

Response. We added **Section VI-G** (“Computational Efficiency and Deployability”) with **Table VI**, reporting parameter count, FLOPs per forward pass, single-window latency (mean of 1000 timed runs after 100 warm-up runs, batch size 1), and peak GPU memory for MIDS and the strongest reproducible baselines on an RTX 4090. MIDS attains 1.147 ms per 100-frame window, 8× below the fastest periodic CAN inter-arrival in our Tesla traces, and a peak memory of 27 MB—both well within the envelope of contemporary automotive gateway ECUs.

Reviewer comment. *In Table 2, the CrySyS CAN Dataset should not be listed under the “simulated” category but under “real & simulated”.*

Response. Corrected. **Table II** now lists CrySyS under “Real & Simulated”, matching the published characterisation of the dataset.

Reviewer comment. *There are also a few minor typesetting errors. For instance, in the first paragraph of the Introduction section, on line 39, a space is missing before the sentence beginning with “However, the protocol’s...”.*

Response. Corrected, together with a careful pass over the manuscript for similar minor issues.

Response to Reviewer 2

We thank Reviewer 2 for the rigorous critique. The reviewer raised seven substantive concerns. All seven have been addressed through the new cross-dataset benchmark (**Table VII**, **Section VI-F**), the explicit disclosure of train/test/window parameters (**Section VI-A5**, **VI-B3**), the formal threat model (**Section IV**), and updated discussion of recent literature including the four references the reviewer suggested.

Reviewer comment. *The paper claims they are able to outperform other models; however, the results shown were tested on other datasets (other than the authors’ own model), as far as I can infer [...]. So these results, unless the authors recreated them using the models on their dataset, are questionable.*

Response. We have made the reproduction methodology explicit in **Section VI-B3**. All baseline numbers in our manuscript—both on our Tesla dataset (**Table IV**) and on the four public datasets (**Table VII**)—are obtained by re-training every baseline from scratch on the target dataset under a single unified protocol: 50 epochs, Adam, $\text{lr} = 10^{-4}$, cosine annealing with $T_{\max} = 10$, gradient clipping at ℓ_2 -norm 1.0, batch size 1024, 5-fold block-shuffled cross-validation, identical input tensor shape ($B, L=100, F=9$) for every model, and identical four-class output head {Normal, ID, Data, Both}. No baseline shares weights with MIDS or borrows numbers from its originating publication. The complete re-implementation is released in our public repository.

Reviewer comment. *There are a number of IDS papers (see [1] to [4] for example), that has superior performance than the SOTA models listed in the paper [...].*

Response. We have integrated all four references into **Section II-A**: Khandelwal and Shreejith (ASAP 2023), Park *et al.* (IEEE Access 2023), Ma *et al.* (Security and Communication Networks 2022), and Rangsikunpum *et al.* (Journal of Systems Architecture 2024) are now discussed as a coherent line of work pursuing FPGA-, graph-classification-, and lightweight-GRU-based detectors. We respectfully note that all four are evaluated exclusively against fabrication-style (injection) threat models, where per-ID inter-arrival statistics provide the primary detection signal. Under the masquerade model adopted in our work (**Section IV**) this signal vanishes by construction, so their headline F1 figures on injection benchmarks are not directly comparable to MIDS’s F1 on masquerade benchmarks; the revised related-work discussion makes this distinction explicit.

Reviewer comment. *[...] the data spoofing attack in the Open CAN dataset has the same signature as the tampering attack in this case, so the performance could be comparable on this dataset.*

Response. This observation is an empirical prediction that our new experiments directly test. On OTIDS (Open CAN-derived benchmark), **Table VII** shows MIDS at $F1 = 99.61\%$ while the strongest baselines (DCNN, CanTransformer) both reach the 100% ceiling, so the ranking is ambiguous at saturation. To provide a sharper comparison, the revision evaluates four public benchmarks; on the harder masquerade-style benchmarks ROAD and CrySyS, MIDS exceeds the strongest baseline by +13.94 and +6.80 percentage points respectively, providing exactly the discriminating evidence the reviewer’s hypothesis requires.

Reviewer comment. *The testing methodology is not clear—are the results reported on the training set [...]? What is your train-validation-test split [...]? Do you also use data balancing for your test set? What is the batch size / event-window size for your test sets?*

Response. We added **Section VI-A5** (“Data Leakage Prevention and Evaluation Protocol”), which states the protocol explicitly: 5-fold block-shuffled chronological cross-validation; in each fold the data are split 80% training / 20% test along contiguous temporal blocks, with no validation split used for tuning (all hyperparameters fixed across folds and listed in **Table III**). Reported metrics are computed exclusively on the held-out 20% test block of each fold and averaged across the five folds. Window length is $L = 100$ frames with stride $S = L = 100$ (non-overlapping);

batch size is 1024 for both training and inference. Dynamic class weighting is applied only to the training loss; the test set retains its natural label distribution.

Reviewer comment. *In your dataset, how do you label an event as an intrusion (at a message level or a block level)? [...] A tampered message requires recalculation of CRC and so on to be considered as a valid message, so your injection setup integrates a malicious ECU?*

Response. Both points are now addressed in **Section IV**. *Labelling level*: attack labels are assigned at the window level (100 frames). *Tampering mechanism and CRC*: the third bullet of **Section IV-A** (“Hardware-assisted CRC”) explains that the attacker has compromised the software stack of an ECU connected to the bus; the on-chip CAN controller of that ECU automatically computes and appends a valid CRC over the tampered payload before transmission, so receiving nodes accept the malicious frames as protocol-compliant. We also clarify in a new paragraph that the attack traces used for evaluation are synthesised *offline* from benign captures, yielding byte-exact ground-truth labels, and that real-time dominant-bit overwriting on a live bus is a separate hardware engineering problem that is not the subject of this detector.

Reviewer comment. *What is the test platform? What is your inference performance (messages per second [...])? What is the energy consumption [...])?*

Response. **Section VI-G** reports the test platform (NVIDIA RTX 4090, batch size 1, 1000 timed runs after 100 warm-up runs) and **Table VI** summarises params, FLOPs, single-window latency, and peak memory for MIDS and the strongest baselines. The MIDS latency of 1.147 ms per 100-frame window corresponds to a throughput of approximately 8.7×10^4 messages/s under windowed inference; the slowest periodic CAN signals in our Tesla traces have inter-arrival periods of ~ 10 ms, leaving an $8\times$ deployability headroom. Energy measurement at the gateway-ECU level requires hardware power-rail instrumentation that is part of our planned follow-up work; we acknowledge this as a limitation in **Section VII** (“Future Works”).

Reviewer comment. *Could you show your model’s performance on other datasets (especially looking at spoofing data) to see if the performance is generalisable?*

Response. This is now the central new evaluation result of the revision. **Table VII** reports F1 for all nine models on all five datasets, including the spoofing-style OTIDS and CT&T datasets specifically requested by the reviewer. MIDS achieves the highest F1 on every dataset, with margins ranging from a near-tie on saturated OTIDS to +13.94 percentage points on ROAD. **Section VI-F** discusses the cross-dataset pattern in detail and observes that MIDS is the only model that ranks first or tied-first under both dense-attack (Tesla, OTIDS, CT&T) and sparse-attack (ROAD, CrySyS) regimes.

Response to Reviewer 3

We thank Reviewer 3 for the careful and constructive feedback and for highlighting both major and minor improvements. The eight major concerns and two minor suggestions are addressed below.

Reviewer comment. *Concern 1. The dataset is organized into 4 folders by driving scenario [...] rather than by the model’s required output classes [...]. This creates a high risk of data leakage [...]. The authors must provide a detailed explanation of their cross-validation strategy that explicitly demonstrates how they prevented this form of data leakage [...].*

Response. We have added two new subsections that jointly address this concern. **Section VI-A4** (“Decoupling Scenario from Attack Label”) explains the two pipeline choices that prevent scenario–label correlation: (i) within-scenario attack replication, where every attack strategy and intensity level is instantiated within every driving scenario, and (ii) cross-scenario interleaving, where per-scenario recordings are concatenated and split into chunks of 1000 windows that are

uniformly shuffled before fold partitioning. **Section VI-A5** (“Data Leakage Prevention and Evaluation Protocol”) details the temporal-leakage safeguards (non-overlapping windows with $S = L = 100$, L -frame safety buffer between adjacent training and testing blocks) and reports an empirical scenario-leakage verification: a held-out classifier trained to predict the driving scenario from a window achieves accuracy indistinguishable from chance after our pipeline, whereas the same classifier achieves $> 97\%$ accuracy on the unshuffled raw recordings. This confirms that scenario information is not a detectable signal in the data presented to MIDS.

Reviewer comment. *Concern 2. The paper is not reproducible in its current form. [...] The authors must provide the fully labeled dataset [...] and add a dedicated “Data Preprocessing” subsection that details the exact transformation from raw logs to model-ready tensors.*

Response. We have made the pipeline fully reproducible. **Section VI-A5** now describes the windowing and labelling transformation (sliding window with $S = L = 100$, window-level majority-vote labelling, block-shuffled fold construction with 1000-window chunks). The full preprocessing code—including the `.asc-to-.csv` conversion script, the attack synthesis tool that produces byte-exact ground-truth labels, the windowing module, and the fold-generation utility—is released in our public repository together with the full labelled dataset in CSV form (one label per window). The repository URL is given in the footnote of **Section I-B**.

Reviewer comment. *Concern 3. The description of the model’s architecture is incomplete. The method for fusing the forward and backward Mamba outputs is described only as a “weighted integration mechanism” [...]. Provide the precise mathematical formulation [...] and update the model architecture diagram (Fig. 3) to explicitly illustrate this mechanism.*

Response. We provide the precise fusion formulation in **Section V-D**: the global representation is $\mathbf{Y}_{\text{global}} = \alpha \cdot \mathbf{H}_{\text{fwd}} + \beta \cdot \mathbf{H}_{\text{bwd}}$, where α and β are scalar learnable parameters jointly optimised with the rest of the network. The fused representation is then flattened and passed through a fully connected layer followed by softmax: $\hat{y} = \text{Softmax}(\mathbf{W}_{\text{out}} \cdot \text{Flatten}(\mathbf{Y}_{\text{global}}) + \mathbf{b}_{\text{out}})$. **Figure 3** (model architecture) has been redrawn to show the two learnable scalars α and β explicitly at the merge point of the forward and backward branches.

Reviewer comment. *Concern 4. The paper fails to demonstrate the generalizability of the MIDS framework. [...] evaluate the MIDS framework on at least one other public benchmark dataset containing tampering attacks, such as the CrySyS dataset.*

Response. The revision evaluates MIDS on *four* public benchmarks: ROAD, CrySyS, OTIDS, and CT&T (**Table VI** and **Section VI-F**). MIDS attains F1 from 93.70% (CT&T) to 99.61% (OTIDS), with 95.76% on the CrySyS benchmark specifically requested by the reviewer. The cross-dataset performance is further contextualised in **Table VII** against eight reproduced baselines under the unified protocol.

Reviewer comment. *Concern 5. The experimental comparison is incomplete. The set of baselines is missing a crucial comparison against modern Transformer-based models [...].*

Response. CanTransformer (Jo and Kim, IEEE Access 2024), a recent Transformer-based IDS, has been added to the baseline set. It now appears in every comparison table (**Table IV** on Tesla and **Table VII** on the cross-dataset benchmark). **Section VI-D** discusses its performance explicitly: CanTransformer is the strongest baseline on Tesla ($F1 = 88.66\%$, second only to MIDS at 96.94%) and ties for the strongest baseline on OTIDS and CT&T, supporting the framing of Mamba versus attention as the key architectural comparison.

Reviewer comment. *Concern 6. The evaluation relies on overly simplistic metrics for a security application. Aggregate scores like F1-score can hide an unacceptably high False Positive Rate [...]. Provide a more detailed analysis [...] specifically reporting the FPR and FNR. Include a discussion on the practical trade-offs, potentially supplemented by a ROC curve.*

Response. We now report FPR and FNR for MIDS on every dataset in **Table VI** (FPR $\leq 1.03\%$ and FNR $\leq 8.49\%$ across all five evaluated benchmarks; the elevated FNR on CT&T reflects its $> 95\%$ -majority Normal class, an artefact of the original release rather than a model limitation). We also include per-class ROC curves and per-class AUC in **Figure 6(c)**, showing that MIDS achieves AUC > 0.998 in every attack class on our Tesla dataset. **Section VI-F** discusses the FPR/FNR trade-off explicitly.

Reviewer comment. *Concern 7. The claim of a “novel” model is an overstatement. [...] Suggestion: Rephrase to use the more accurate term “innovative framework”.*

Response. We have adopted the reviewer’s suggested phrasing throughout the manuscript. Architectural claims now use “innovative framework”; the term “novel” is reserved for contributions that are genuinely new, namely the publicly released Tesla dataset and the first-time application of Mamba to CAN-bus masquerade detection.

Reviewer comment. *Concern 8. The description of the training and testing protocol lacks explicit detail. The paper mentions a “5-fold cross-validation” approach but does not explicitly state the train/test split ratio (e.g., 80/20) [...].*

Response. **Table III** now lists “Train/Test split: 4:1” (i.e., 80%/20%) explicitly. **Section VI-A5** adds a sentence stating: “In each fold, four blocks (80%) form the training set and the remaining block (20%) is held out for testing, so the test traffic is always chronologically distinct from the training traffic.”

Minor suggestions.

Reviewer comment. *Minor 1. [...] the models listed (Attention, CNN, LSTM, etc.) are architectures, not specific published methods. It would be stronger to cite the specific papers from which these results were reproduced for clarity as some references are missing for KNN, Attention and MLP.*

Response. We have replaced the generic architecture labels in **Table IV** with specific published method names and citations (GIDS [Seo *et al.* PST 2018], CANShield [Shahriar *et al.* IoT-J 2023], CanBus-IDS [Hoang and Kim, Veh. Commun. 2022], DCNN [Song *et al.* Veh. Commun. 2020], CANTransfer [Tariq *et al.* SAC 2020], CanTransformer [Jo and Kim, IEEE Access 2024]), and folded the foundational MLP/CNN into a separately labelled block. **Table VII** maintains the same convention.

Reviewer comment. *Minor 2. The term “spoofing” is used in the caption for Fig. 7, but the rest of the paper uses “tampering”. Please ensure consistent terminology throughout.*

Response. Corrected. Figure 7’s caption and all related class labels now use “ID tampering”, “data tampering”, and “combined tampering” consistently with the rest of the manuscript.

We are grateful to all three reviewers for the depth of their engagement with the manuscript. The cross-dataset benchmark, the formal threat model, and the explicit reproduction protocol introduced in this revision were directly motivated by their comments and substantially strengthen the paper. We hope that the changes meet the reviewers’ expectations.

Sincerely,
The Authors